ICKG 2026

Griffith UNIVERSITY

Graph-Based Personalized Memory for LLM Agents

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Contributions

01 A lifecycle taxonomy of Graph-based Personalised Memory for agents, covering memory representation, evolution, retrieval and evaluation.

Representation Evolution Retrieval Evaluation

02 Benchmark summary.

03 Timely review.

04 Open research directions.

Disclosure: presentation content is fully written by the presenter, final layout is edited by AI.

Background

LLM agents are serving as personal assistants

 OpenClaw  Hermes  Grok Bot

→ Their recommendations and responses are shaped by human behaviours.

LLM context windows are finite

→ Storing all interactions in the context window is impractical

Memory is mostly stored as text files or vector stores.

→ Memory relationships, evolution, deletion are not explicitly expressed.



Knowledge Graph (KG) is a good candidate to address these limitations while keeping the memory interpretable and auditable.

MEMORY REPRESENTATION

Memory elements — Node

Evidence node

original interaction records with little to no format modification.

E.g. “Tom went to Shenyang for the ICKG conference and met a lot of experts in his field”

Fact node

extracted atomic assertion or entity.

E.g. “Tom went to Shenyang” or “Tom”

Abstraction node

synthesized summary observations that form higher-level user models.

E.g. “Tom’s China trip for ICKG conference”

Domain-specific node

dedicated to storing entities about one particular topic, such as sports, prescriptions or family members.

These memory nodes are only categorized by function. The specific content or format inside the nodes can vary vastly.

MEMORY REPRESENTATION

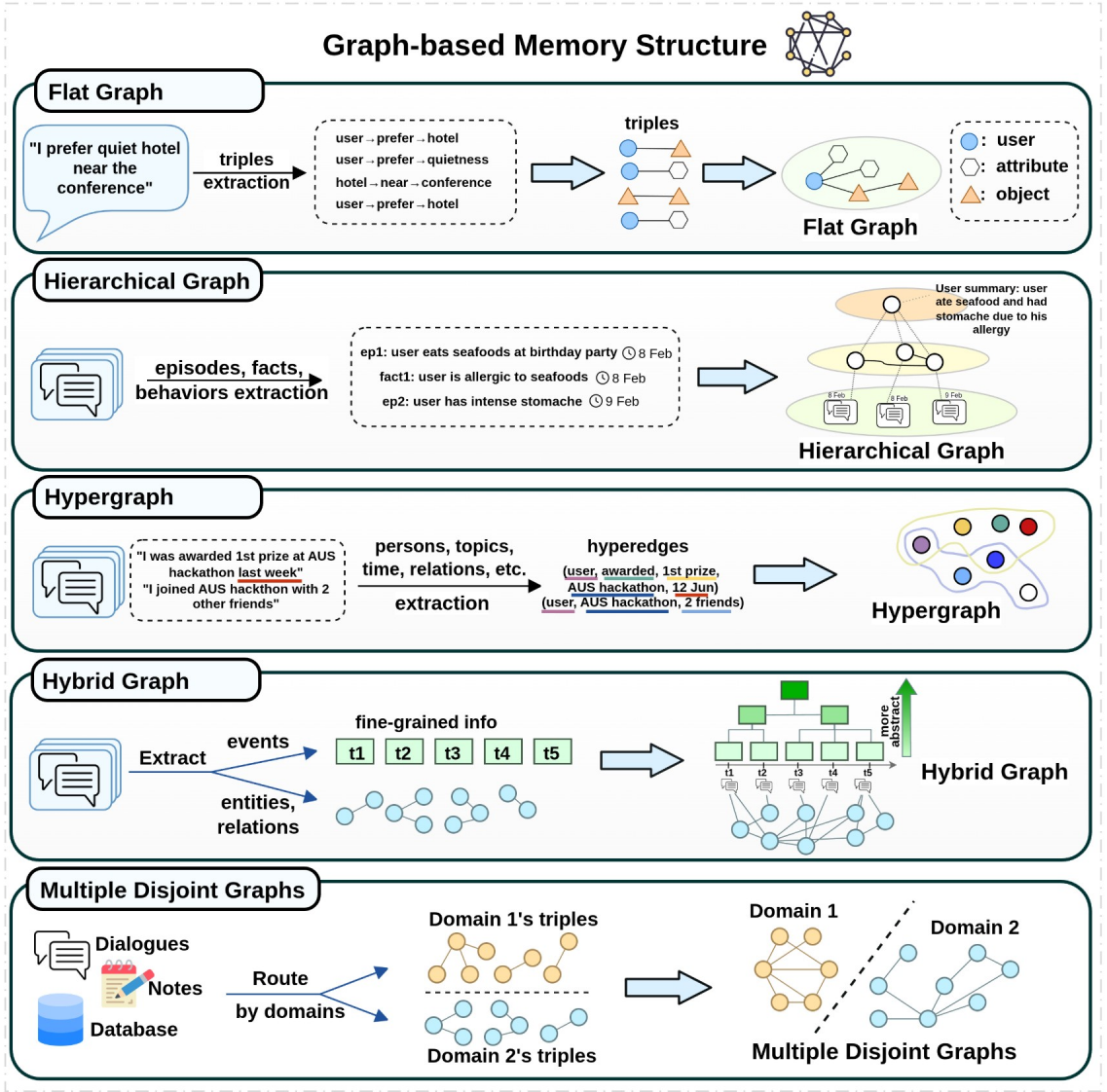
Memory elements — Relation

Temporal	encodes event order or progression. E.g. “Tom went to Shenyang” → “Tom met a lot of experts in his field”
Provenance	connects assertions or abstractions to source evidence. E.g. “I just finished my 2-day ICKG conference. There are so many talented people at ICKG, mate.” – Tom said -evidence_for→ “Tom attended 2-day ICKG conference”, “Tom met many experts”
Associative	links contextually or semantically related memories. E.g. “Tom gave a presentation on Knowledge Graph” ↔ “Tom discussed with Tian how to improve Knowledge Graph memory efficiency”
Domain-specific	encodes task-specific relationships E.g. (care assistant) “Nurse Anna” -supervises→ “Tom: 8am medication”

MEMORY REPRESENTATION

Memory structures

1. Flat graph
2. Hierarchical graph
3. Hypergraph
4. Hybrid graph
5. Multiple distinct graphs

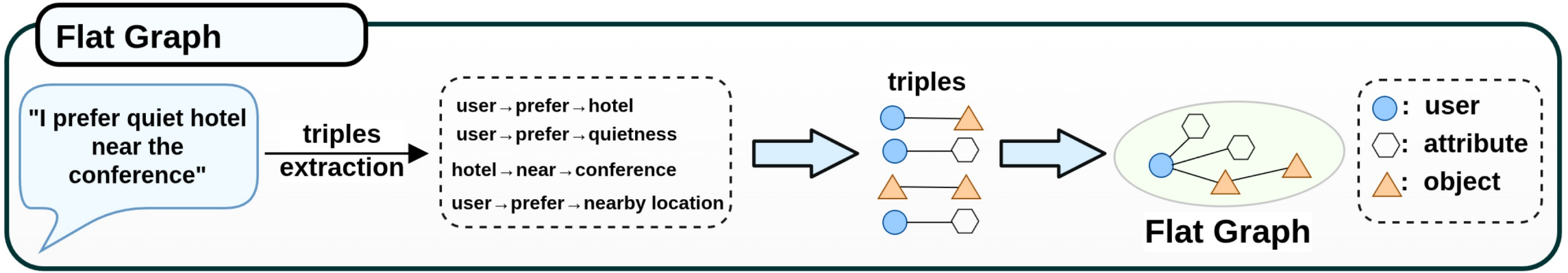


MEMORY STRUCTURES

Flat graph

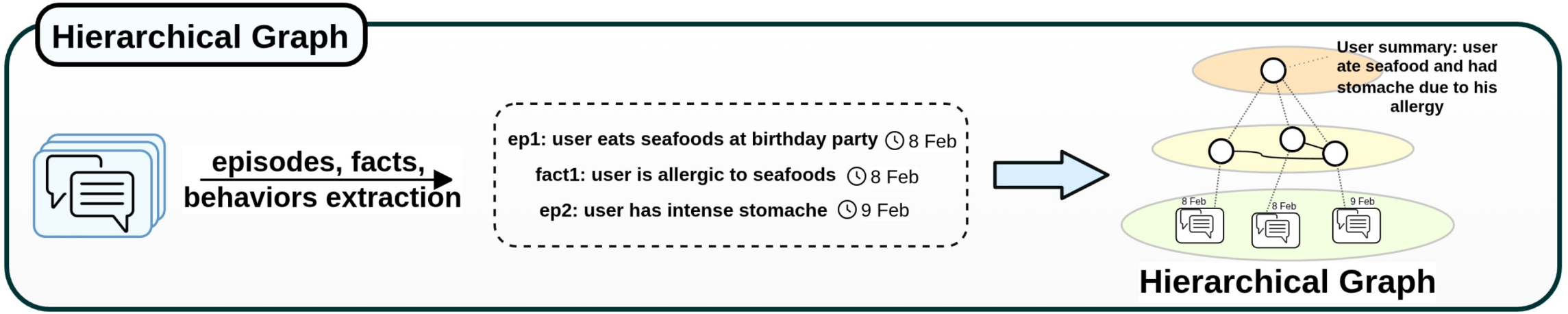
1 Flat | 2 Hierarchical | 3 Hyper | 4 Hybrid | 5 Multiple

> Flat Graph stores facts, entities, preferences and relations without explicit abstraction layers or levels



Hierarchical graph

> Hierarchical graph organizes user memories into multiple layers at different levels of specificity.

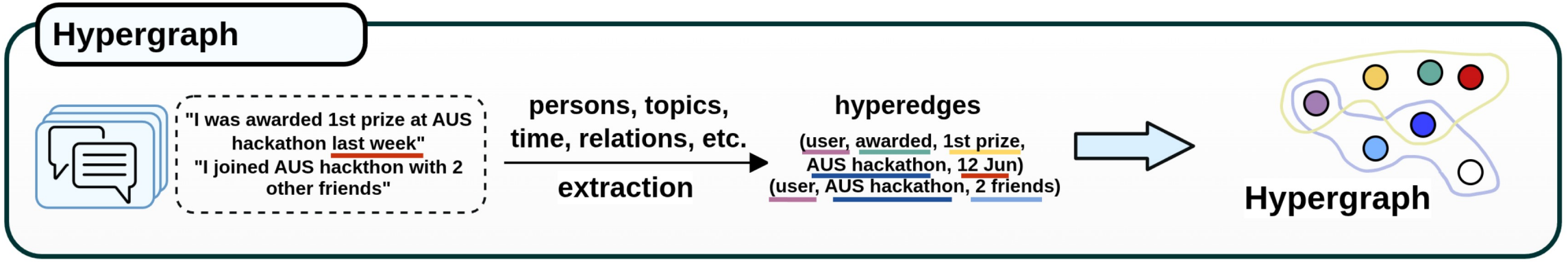


MEMORY STRUCTURES

Hypergraph

1 Flat | 2 Hierarchical | **3 Hyper** | 4 Hybrid | 5 Multiple

> Hypergraph preserves joint context whose meaning depends on several connected elements.

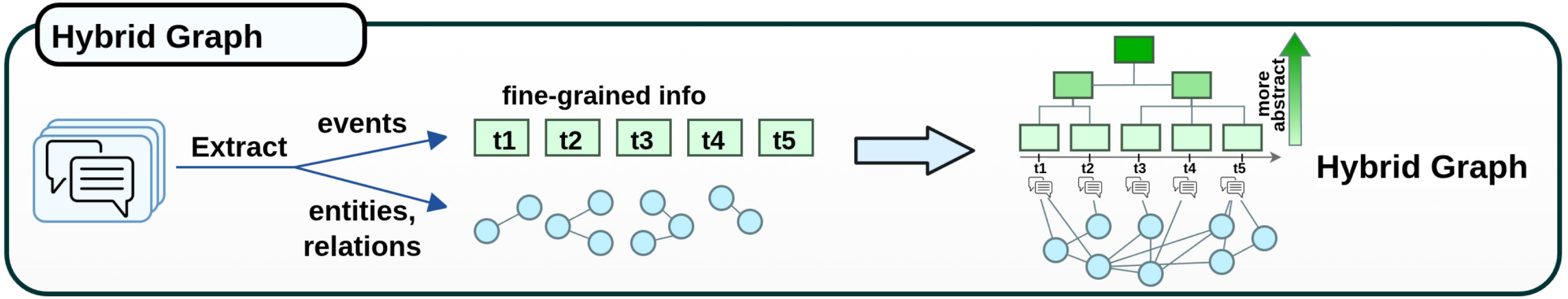


MEMORY STRUCTURES

Hybrid graph

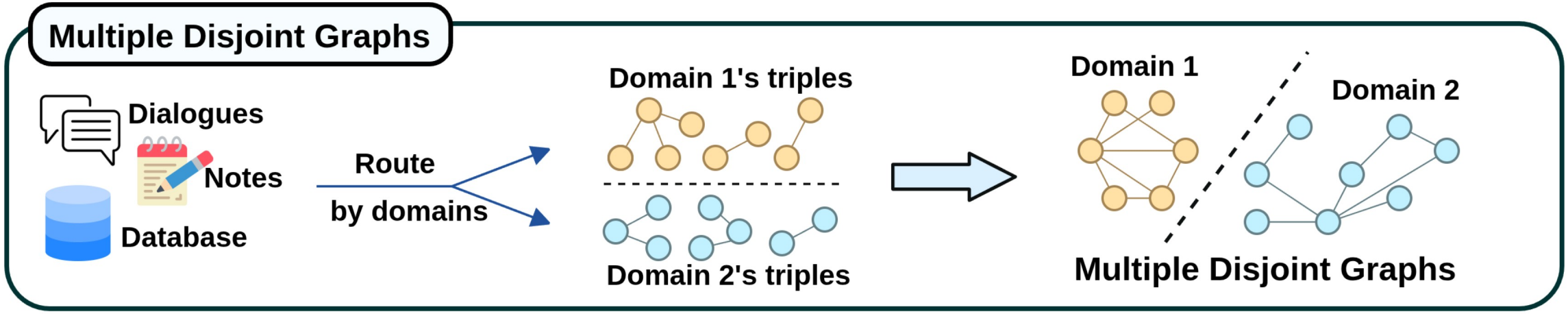
1 Flat | 2 Hierarchical | 3 Hyper | **4 Hybrid** | 5 Multiple

> Hybrid Graph distributes the memories across graph topology and a complementary non-graph substrate such as a tree



Multiple distinct graphs

> Multiple distinct graphs split user memories across separate graph instances, each of which serves a distinct, specialized function



Discussion

Memory representation determines what evidence is retained and how later lifecycle operations use it.

Evidence nodes takes up a lot of storage.

→ Does the use case require provenance?

Abstraction-centered designs compact user memories but may discard helpful fine-grained information

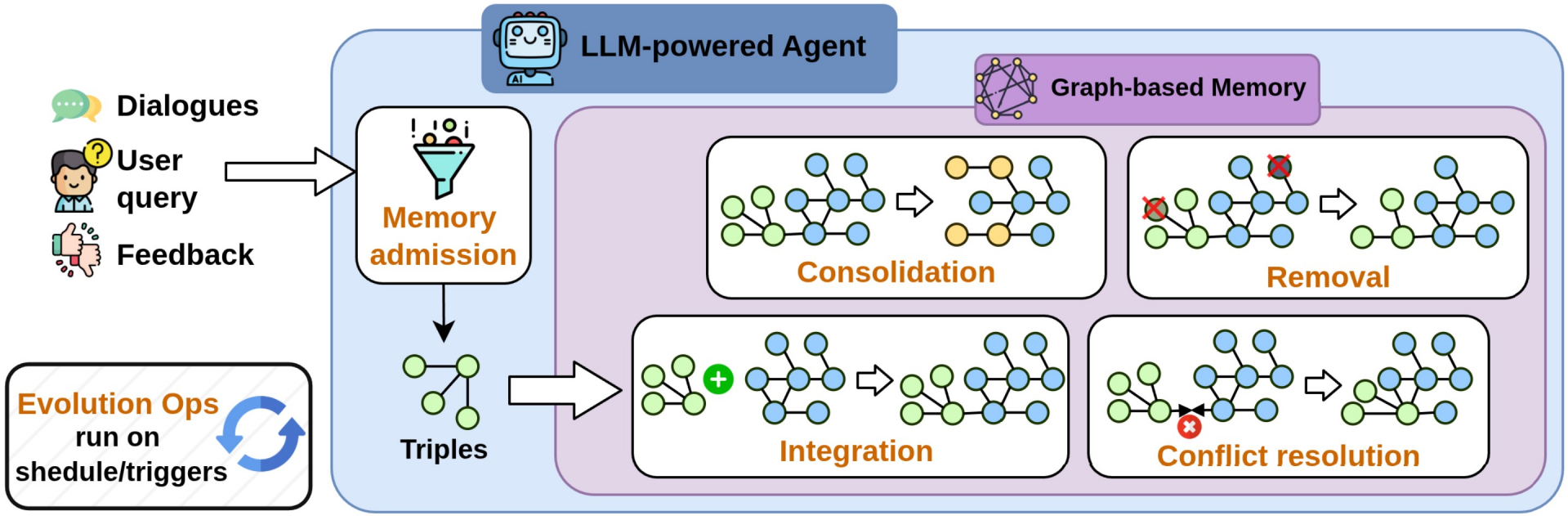
→ A good balance is needed.

Hierarchical and hybrid graphs require information consistency across levels

→ Additional burden to propagate changes

Memory evolution

> maintains an up-to-date memory systems



MEMORY EVOLUTION

Memory admission

1 Admission | 2 Integration | 3 Conflict | 4 Consolidation | 5 Removal

Decides whether incoming evidence should be admitted to become persistent memory.

e.g. retain user-persona-relevant content

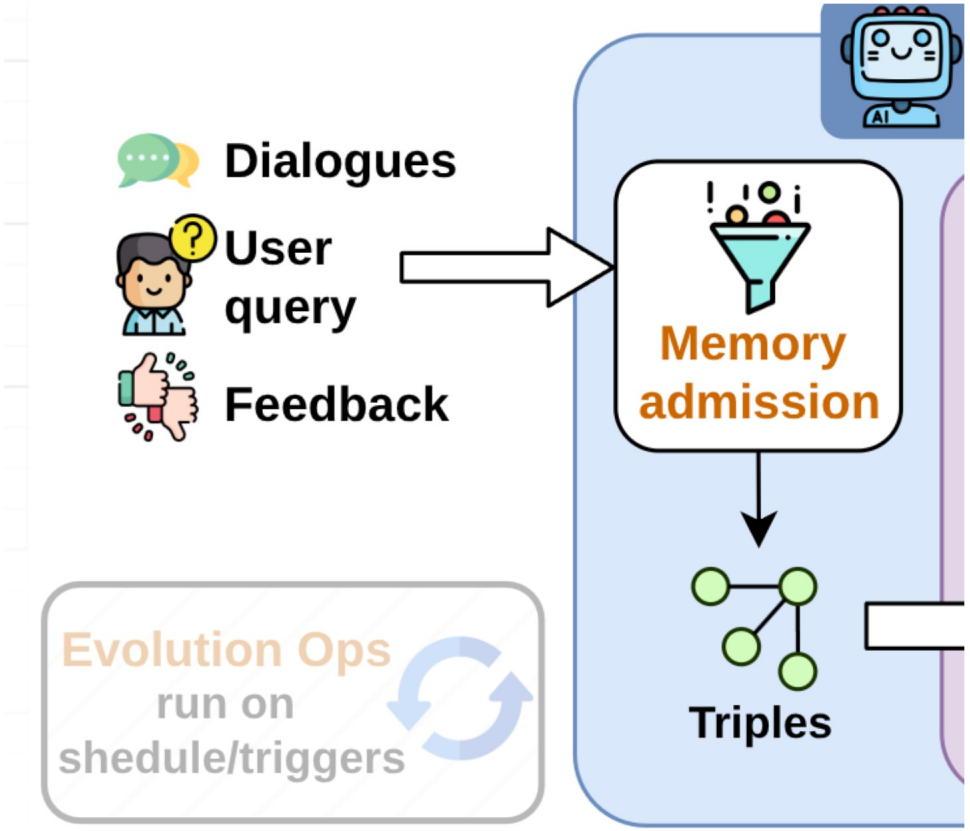
Can be triggered by user feedback, policy, etc.

Over-admission

risks polluting memory with transient information

Under-admission

may discard useful information

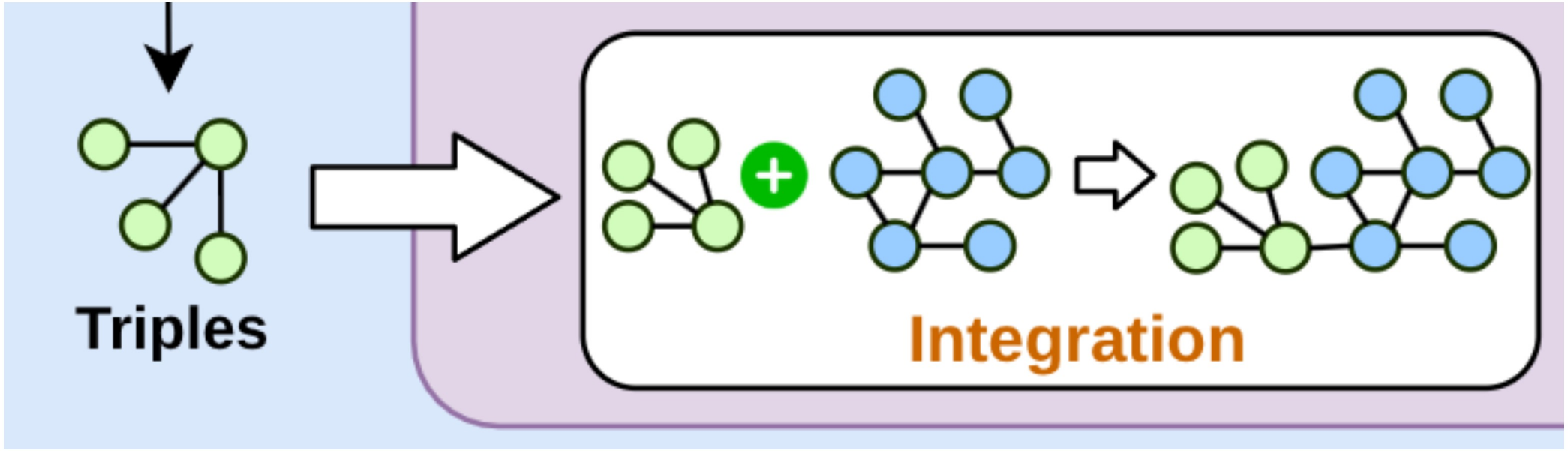


MEMORY EVOLUTION

Memory integration

1 Admission | 2 Integration | 3 Conflict | 4 Consolidation | 5 Removal

> Attaches evidence to existing graph structure

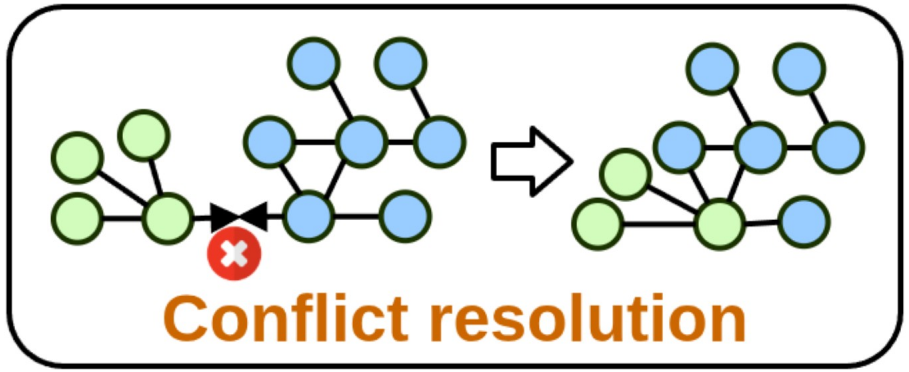


MEMORY EVOLUTION

Conflict resolution

1 Admission | 2 Integration | 3 Conflict | 4 Consolidation | 5 Removal

> Needed when incoming evidence changes the validity of a stored memory



3 CONFLICT TYPES

1

Temporal supersession

2

User correction

3

Contextual coexistence

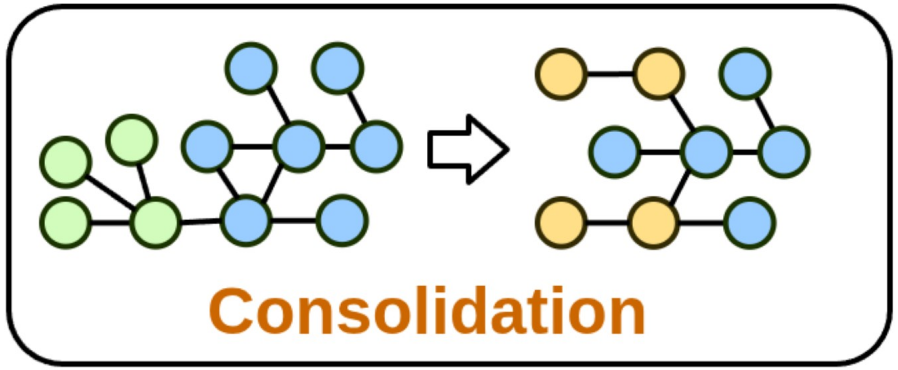
valid in different contexts.

MEMORY EVOLUTION

Consolidation

1 Admission | 2 Integration | 3 Conflict | 4 Consolidation | 5 Removal

> Reorganizes and merges accumulated memory after multiple writes.



SERVES AS

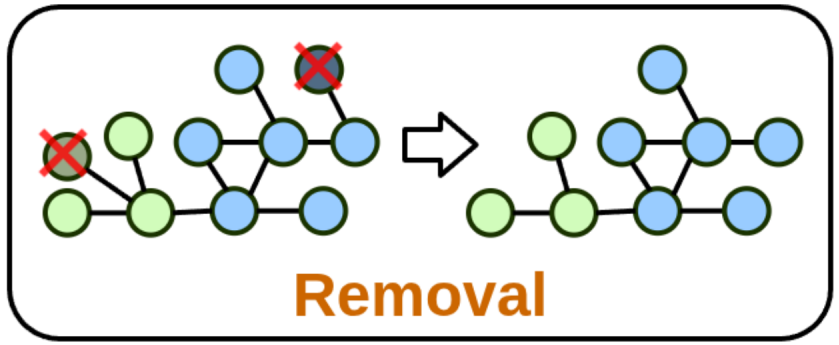
deduplication	summarization	restructuring	archival

functions

MEMORY EVOLUTION

Memory removal

1 Admission | 2 Integration | 3 Conflict | 4 Consolidation | 5 Removal



> Takes 2 distinct forms: Forgetting and Deleting

Forgetting

reduces a memory's exposure, resolution, weight or retrieval likelihood while memory content stays unchanged

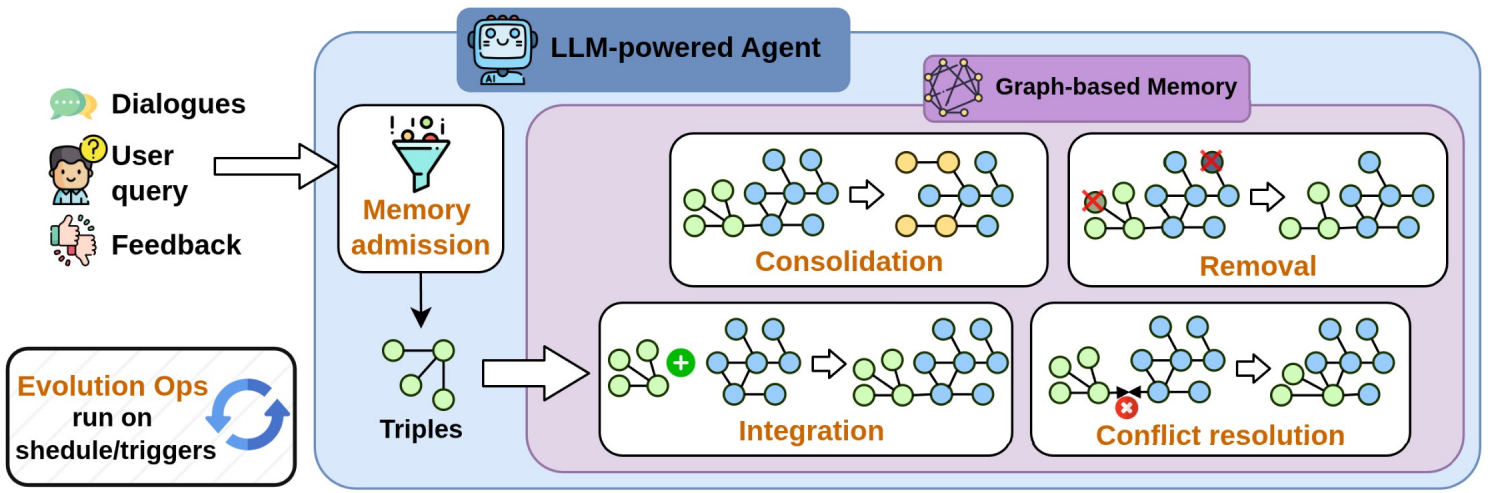
Deleting

either removes memories completely or deactivates them (i.e. soft deletion), excluding them from retrieval unless explicitly requested.

Discussion

These operations often happen simutaneously.

Must be done with care as memories are tightly connected.



Memory retrieval

> maps a user request to a bounded subset of memories.

3 MECHANISMS

1

Similarity-based retrieval

2

Structure-based retrieval

3

Agentic retrieval

They can be used individually or together.

MEMORY RETRIEVAL

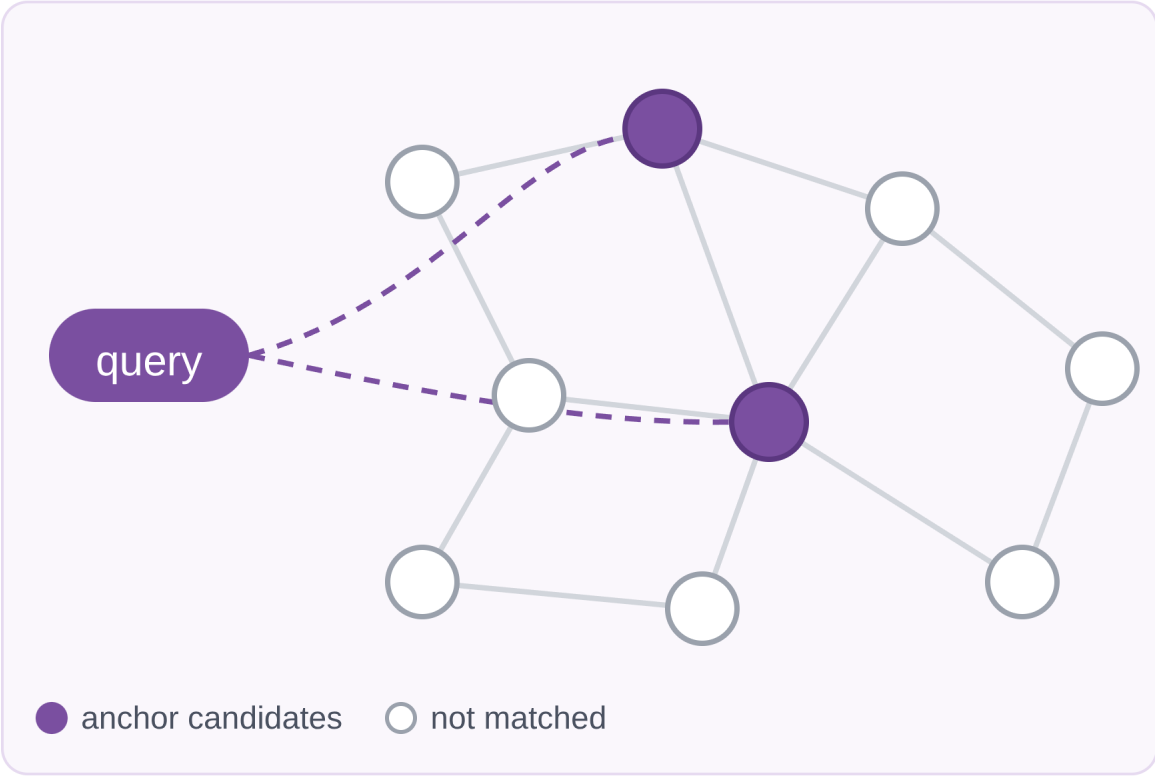
Similarity-based retrieval

1 Similarity | 2 Structure | 3 Agentic

lexical / keyword / dense-vector matching.

commonly used for generating anchor candidates to expand in the KG.

> Searching based on semantic proximity alone may miss evidence connected through temporal, causal, entity or abstraction relations.



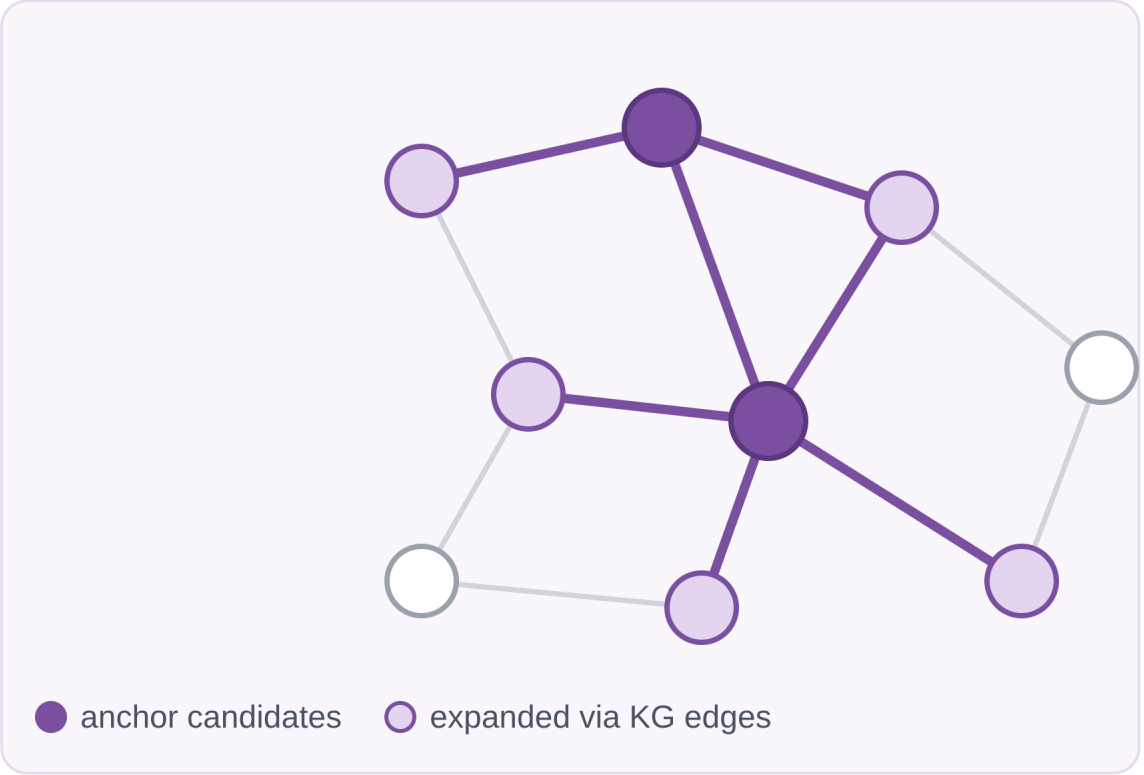
MEMORY RETRIEVAL

Structure-based retrieval

1 Similarity | 2 Structure | 3 Agentic

leverages graph structure to expand initial candidates via KG edges.

The expansion can complement the missing evidence provided by similarity-based retrieval.

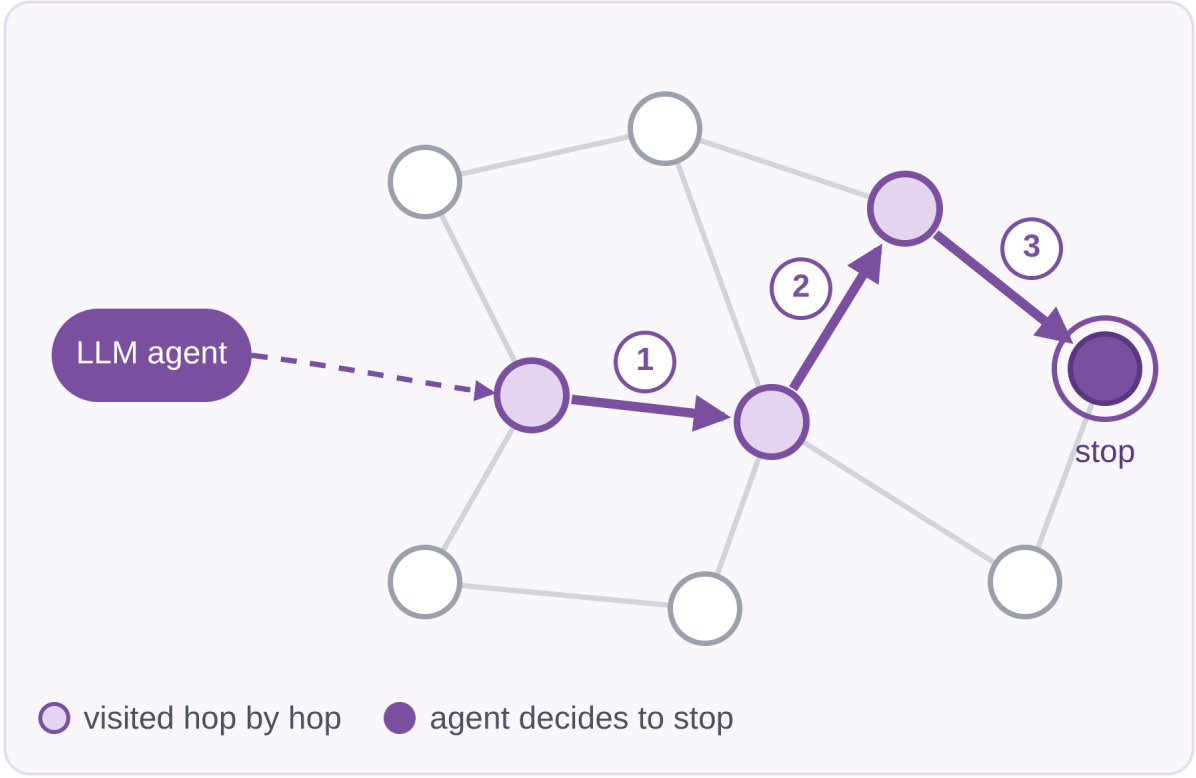


MEMORY RETRIEVAL

Agentic retrieval

1 Similarity | 2 Structure | 3 Agentic

- leverages LLM intelligence to interpret the user query.
- steers graph exploration, stopping and routing dynamically based on intermediate evidence.



Discussion

The 3 mechanisms can be combined in a retrieval pipeline.

Similarity + Structure + Agentic

The 3 memory lifecycle operations (retrieval strategy, evolution and graph structure) should be co-designed since they are highly dependent on each other.

EVALUATION

Evaluation

Current benchmarks focus on downstream performance:

answer correctness | supporting-evidence identification | inference cost | user modelling state

They fall into 3 groups:

Group	Benchmarks
Conversational, long-horizon recall and retrieval	LoCoMo, LongMemEval
Personalization and user-state modeling	PersonaMem, PersonaMem-v2, RealMem, PerLTQA
Graph structure and evolution	EngramaBench, EvoMemBench, EvoArena, ActMemEval, ATM-Bench, StructMemEval

Discussion

No part of the personalized memory lifecycle is more important than the rest. Every part should be co-designed.

→ This can be a challenge at the design stage, or an opportunity to re-think memory systems.

Can it scale for lifelong personal memory (with up to billions of interaction tokens)?

Can it support multimodal input?

AUTHOR CORNER

Observations/insights since publishing this survey

Nodes

can also be structure- or format-rich text. The same goes for edges.

E.g. MemCog [1] has a semi-graph structure with hyperlinks connecting one text file to another (think Obsidian) → more coherent memory

Scalability

O(n)
current memory systems

O(1)
human memory

Current memory systems' storage scales approximately O(n) with input data. Human memory scales O(1) with input data – a reasonable baseline?

E.g. MeMo [2] offers a glimpse of this by training a small language model to serve as memory.

[1] Z. Li, X. Fan, F. Li, and W. Que, "MemCog: From memory-as-tool to memory-as-cognition in conversational agents," 2026, arXiv:2605.28046.
[2] R. W. H. Quek *et al.*, "MeMo: Memory as a model," in *Proc. 43rd Int. Conf. Mach. Learn. (ICML)*, 2026.

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Observations/insights since publishing this survey

There are still gaps in other domains with their unique challenges.

E.g. medical – MedMemoryBench [3]

Towards portable, specialized LLMs: essential in the medical sector with its sensitive data.

RAW-RAG [4] challenges existing approaches with graph-based memory systems.

Do knowledge graphs provide superior performance or address your use case’s pain points (e.g. provenance)?

[3] Y. Wang *et al.*, “MedMemoryBench: Benchmarking agent memory in personalized healthcare,” in *Adv. Neural Inf. Process. Syst. (NeurIPS), Datasets and Benchmarks Track*, 2026.

[4] Z. Bao, S. Zhang, H. Zhang, S. Lou, M. Wen, and M. Luo, “Is complex agentic memory always needed? Naive RAG is stronger than we thought,” 2026, arXiv:2610.30755.

Thanks for listening!

All relevant resources:
rodonguyen.dev/ickg26



Disclosure: presentation content is fully written by the presenter, final layout is edited by AI.